



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

GEOGRAPHY
PAPER 1

6037/1

JUNE 2023 SESSION

3 hours

Additional materials:
Answer paper

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There are **ten** questions in this paper.

Answer any **four** questions in this paper.

Write your answers on the separate answer paper provided.
If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
Sketch maps and diagrams should be drawn wherever they serve to illustrate an answer.
You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 6 printed pages and 2 blank pages.

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Answer any four questions.

- 1 (a) Explain the term *soil catena*.

- (b) Figure 1.1 shows the global distribution of vegetation belts. [6]

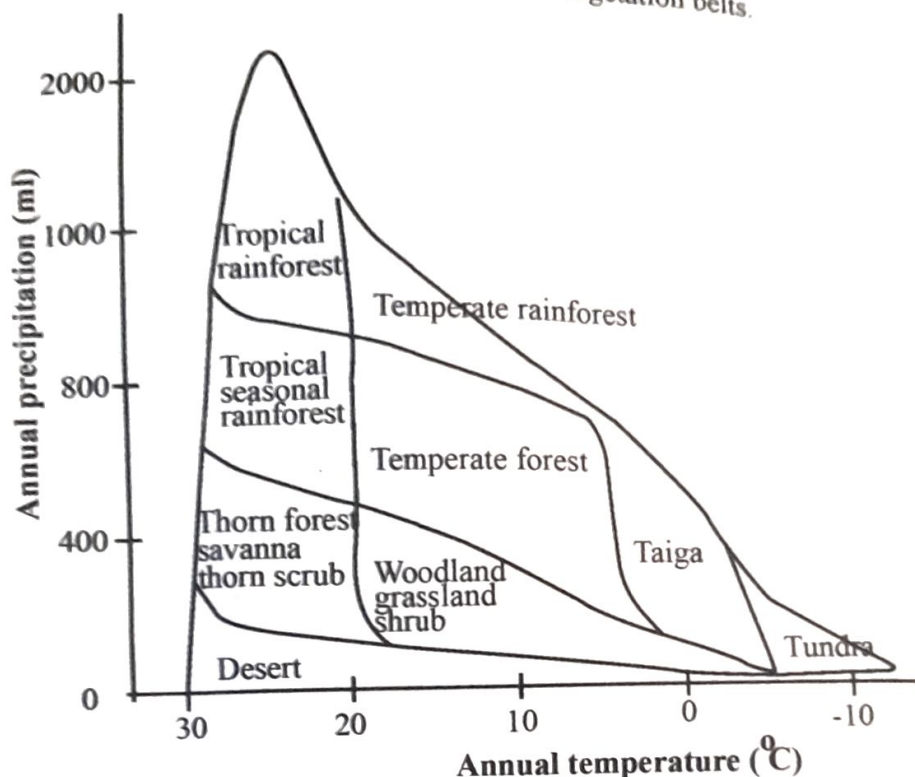


Fig. 1.1

Explain the distribution of tropical rainforests and grasslands shown. [12]

- (c) With reference to a named area, assess the effectiveness of measures which are introduced to protect tropical grassland areas. [7]

- 2 (a) Explain the terms *biomass* and *community* as used in the study of vegetation. [6]

- (b) Discuss ways in which natural vegetation of the tropical savannah has been modified by biotic factors. [12]

- (c) Assess the impact of human activities in the modification of the natural vegetation in tropical areas. [7]

- 3 (a) Explain how rock joints are formed. [6]

- (b) Show how rock type and bedding planes may influence weathering and land forms in limestone regions. [12]

- (c) Discuss the roles of vegetation, relief and human activities in weathering. [7]

- 4 (a) Define the terms *regolith* and *scree*. [6]

- (b) Figure 4.1 shows variations in the depth of weathering for climate zones from the poles to the tropics.

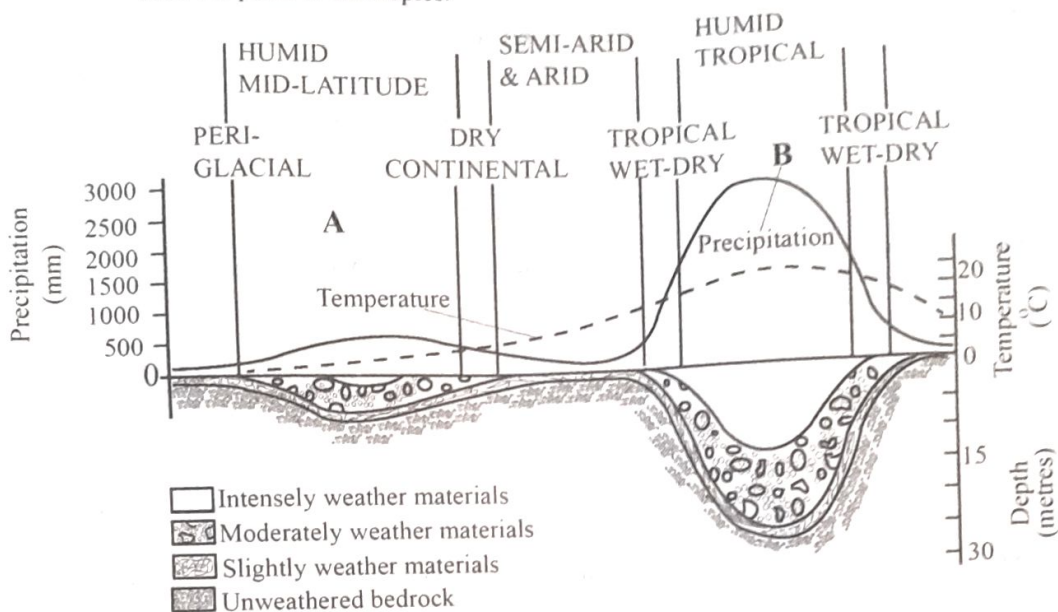


Fig. 4.1

Explain the conditions that are responsible for the differences in regolith depths in the areas marked A and B. [12]

- (c) To what extent can the reduction in regolith depths in some areas be attributed to human activities? [7]

- 5 (a) Figure 5.1 shows the global distribution of tectonic plates.

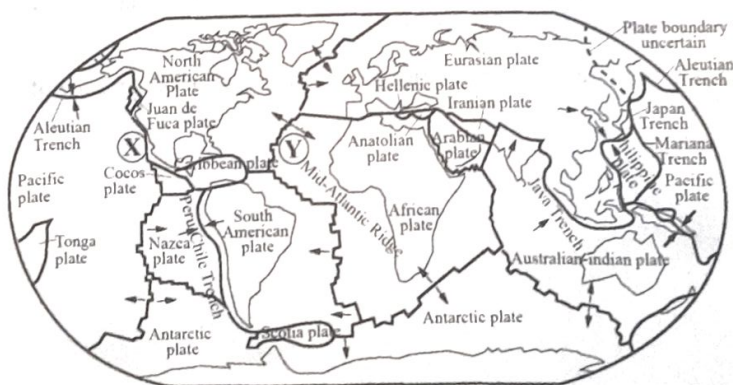


Fig. 5.1

Describe the formation of plates as shown in Fig. 5.1 above. [6]

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- (b) With the aid of diagrams explain the processes operating at boundaries X and Y. [12]
- (c) Explain the development of any **two** landforms associated with intrusive volcanic activity. [7]
- 6 (a) Describe the sources of the sediment load of a river. [6]
- (b) Use diagrams to explain the development of the following river channel landforms:
- (i) river cliff,
 - (ii) flood plain,
 - (iii) delta. [12]
- (c) Explain how human activities can modify channel features. [7]
- 7 (a) Define the terms *interception* and *infiltration*. [6]
- (b) Use diagrams to explain how the global water balance is achieved. [12]
- (c) How have human activities affected the water balance within a drainage basin you have studied? [7]
- 8 (a) Define the terms *ephemeral*, *intermittent* and *perennial* as used in the study of streams. [6]
- (b) Explain the factors which influence variations in river discharge from place to place and from time to time. [12]
- (c) To what extent are human activities responsible for the occurrence of droughts? [7]

- 9 (a) Draw a labelled diagram to illustrate how the global energy balance is achieved. 6]
- (b) Photograph below shows a tropical storm. Figure 9 shows variations in pressure and wind speed from the centre of the storm.

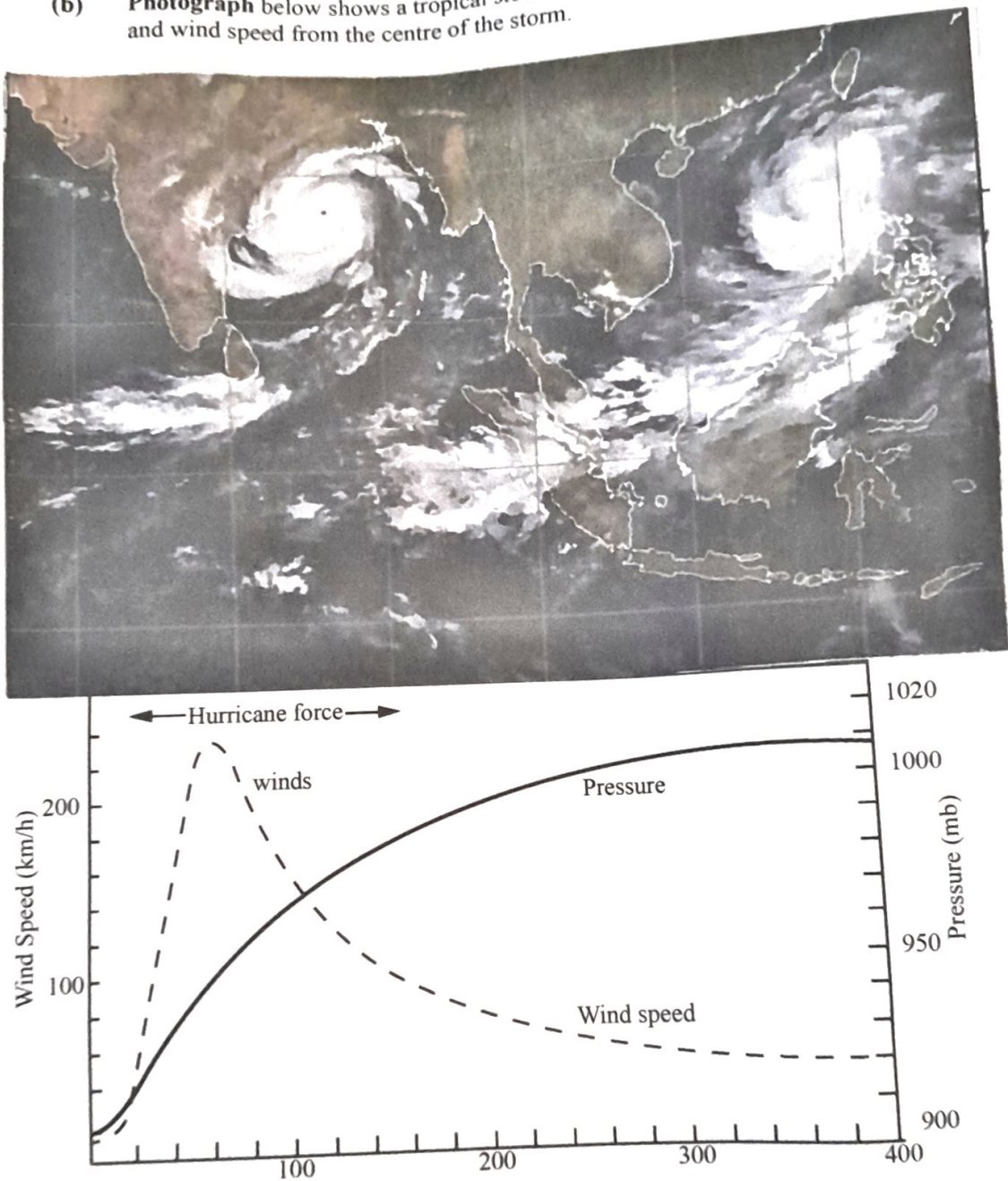


Fig. 9

- (i) Describe the conditions that favour the development of such storms.
- (ii) Explain the variations in wind speed and precipitation across the storm.

[12]

- 10
- (c) Use examples to evaluate the effectiveness of any two measures taken to reduce the adverse impacts of tropical storms. [7]
 - (a) Describe the Bergeron-Findeisen process of raindrop formation. [6]
 - (b) Explain how clouds are classified. [12]
 - (c) Outline conditions under which temperature inversions occur. [7]